



Siemens Matsushita Components

SAW Components Low Loss Filter

B4812
246,01 MHz

Data Sheet

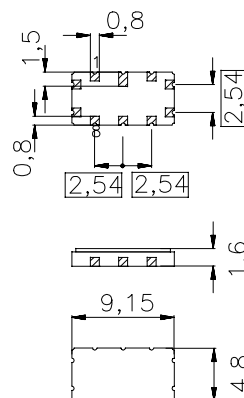
Ceramic package **QCC10B**

Features

- Low-loss IF filter for mobile telephone
- Channel selection in GSM systems
- Hermetically sealed ceramic SMD package
- Balanced and unbalanced operation possible

Terminals

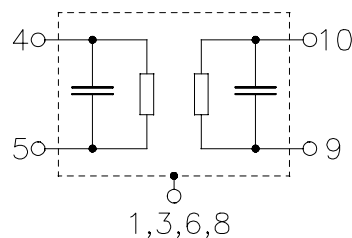
- Gold-plated Ni



Dimensions in mm, approx. weight 0,23 g

Pin configuration

4	Input
5	Input ground or balanced input
9	Output
10	Output ground or balanced output
1, 3, 6, 8	Case – ground
2, 7	Ground



Type	Ordering code	Marking and Package according to	Packing according to
B4812	B39251-B4812-Z710	C61157-A7-A49	F61074-V8035-Z000

Electrostatic Sensitive Device (ESD)

Maximum ratings

Operable temperature range	T	- 25/+ 85	°C
Storage temperature range	T_{stg}	- 25/+ 85	°C
DC voltage	V_{DC}	0	V
Source power	P_s	10	dBm



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Characteristics

Reference temperature:

$$T = 25^{\circ}\text{C}$$

Terminating source impedance:

$$Z_S = 700\ \Omega \parallel -2,6\ \text{pF}$$

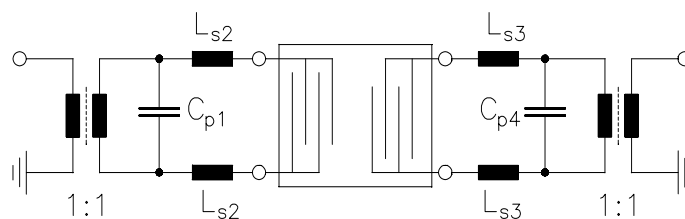
Terminating load impedance:

$$Z_L = 700\ \Omega \parallel -2,6\ \text{pF}$$

		min.	typ.	max.	
Nominal frequency	f_N	—	246,01	—	MHz
Minimum insertion attenuation (including loss in matching coils)	$\alpha_{\min}$	2,0	3,2	5,0	dB
Amplitude ripple (p-p)	$\Delta\alpha$				
$f_N - 67,5\ \text{kHz} \quad \dots \quad f_N + 67,5\ \text{kHz}$		—	0,6	2,0	dB
$f_N - 80,0\ \text{kHz} \quad \dots \quad f_N + 80,0\ \text{kHz}$		—	0,7	3,0	dB
Group delay ripple (p-p)	$\Delta\tau$				
$f_N - 50,0\ \text{kHz} \quad \dots \quad f_N + 50,0\ \text{kHz}$		—	0,5	1,5	μs
$f_N - 80,0\ \text{kHz} \quad \dots \quad f_N + 80,0\ \text{kHz}$		—	1,2	3,0	μs
Relative attenuation (relative to $\alpha_{\min}$)	α_{rel}				
$f_N - 25,00\ \text{MHz} \quad \dots \quad f_N - 3,00\ \text{MHz}$		50	60	—	dB
$f_N - 3,00\ \text{MHz} \quad \dots \quad f_N - 1,60\ \text{MHz}$		48	60	—	dB
$f_N - 1,60\ \text{MHz} \quad \dots \quad f_N - 0,60\ \text{MHz}$		38	50	—	dB
$f_N - 0,60\ \text{MHz} \quad \dots \quad f_N - 0,40\ \text{MHz}$		28	40	—	dB
$f_N - 0,40\ \text{MHz} \quad \dots \quad f_N - 0,20\ \text{MHz}$		8	14	—	dB
$f_N + 0,20\ \text{MHz} \quad \dots \quad f_N + 0,40\ \text{MHz}$		8	14	—	dB
$f_N + 0,40\ \text{MHz} \quad \dots \quad f_N + 0,60\ \text{MHz}$		28	40	—	dB
$f_N + 0,60\ \text{MHz} \quad \dots \quad f_N + 1,60\ \text{MHz}$		38	50	—	dB
$f_N + 1,60\ \text{MHz} \quad \dots \quad f_N + 3,00\ \text{MHz}$		48	60	—	dB
$f_N + 3,00\ \text{MHz} \quad \dots \quad f_N + 25,00\ \text{MHz}$		50	60	—	dB
Impedance at f_N					
Input: $Z_{\text{IN}} = R_{\text{IN}} \parallel C_{\text{IN}}$		—	700 $\parallel$ 2,6	—	$\Omega \parallel \text{pF}$
Output: $Z_{\text{OUT}} = R_{\text{OUT}} \parallel C_{\text{OUT}}$		—	700 $\parallel$ 2,6	—	$\Omega \parallel \text{pF}$
Temperature coefficient of frequency ¹⁾	TC_f	—	- 0,036	—	ppm/K ²
Frequency inversion point	T_0	—	25	—	$^{\circ}\text{C}$

¹⁾ Temperature dependence of f_c : $f_c(T) = f_c(T_0)(1 + TC_f(T - T_0)^2)$

Test matching network to 50 Ω (element values depend on PCB layout):



$$\begin{aligned} C_{p1} &= 1,8\ \text{pF} \\ L_{s2} &= 56\ \text{nH} \\ L_{s3} &= 56\ \text{nH} \\ C_{p4} &= 1,8\ \text{pF} \end{aligned}$$

Preliminary format of data sheet

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OFW E MF
Dec. 11, 1997



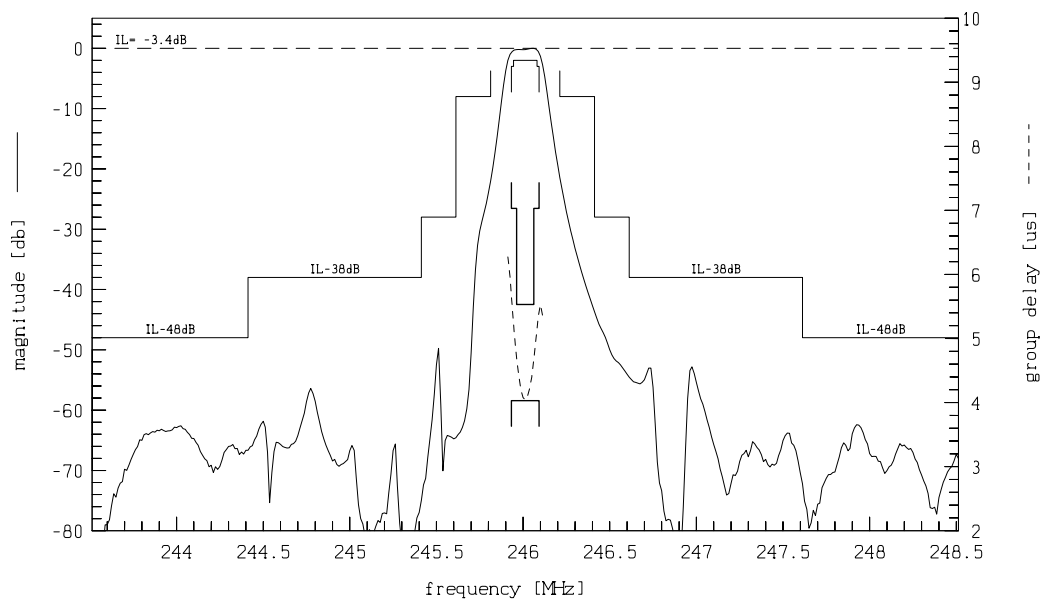
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Transfer function:



Transfer function (pass band):

